WORKED SOLUTION

STEP III 1998, Question 7

Pure • Curve Sketching and Integrals

STEP III 1998, Question 7

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/98-s3-q7

The key idea

The first function is not a random warm-up: when we differentiate $\mathrm{b}(x)$ and set the derivative to zero, the equation that appears is exactly $\mathrm{f}(x/T) = 0$. So everything we learn about f (one positive root, lying between 2 and 3) tells us where b turns. The dependence of B on T comes from the natural substitution $x = Tu$, which strips T out of the integrand.

Solution

Sketching f and finding m

We have $\mathrm{f}(0) = 1 \cdot (-3) + 3 = 0$ and

$$\mathrm{f}'(s) = \mathrm{e}^s(s-3) + \mathrm{e}^s = \mathrm{e}^s(s-2).$$

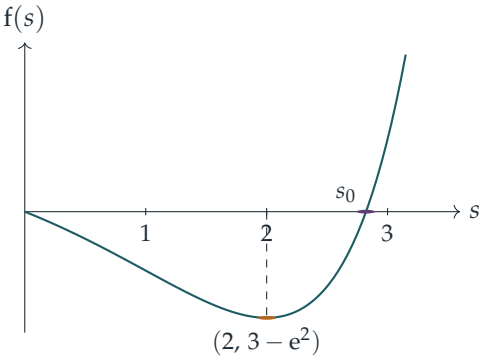
Since $\mathrm{e}^s > 0$, f is decreasing on $[0,2]$ and increasing on $[2,\infty)$, with a minimum $\mathrm{f}(2) = 3 - \mathrm{e}^2 \approx 3 - 7.29 = -4.29$. As $s \to \infty$, $\mathrm{e}^s(s-3) \to \infty$, so $\mathrm{f}(s) \to \infty$.

Because $\mathrm{f}(0) = 0$ and f is strictly decreasing on $[0,2]$, we have $\mathrm{f}(s) < 0$ for $0 < s \leq 2$. On $[2,\infty)$ it is strictly increasing from a negative value to $+\infty$, so it crosses zero exactly once there, at some $s_0 > 2$. Hence s_0 is the *only* positive root of f.

With $\mathrm{e} \approx 2.7$:

$$\mathrm{f}(1) = 3 - 2\mathrm{e} \approx -2.4, \qquad \mathrm{f}(2) \approx -4.29, \qquad \mathrm{f}(3) = 0 + 3 = 3.$$

So the smallest positive integer with $\mathrm{f}(m) > 0$ is $m = 3$, and $2 < s_0 < 3$ (in fact $s_0 \approx 2.82$).



The single turning point of b

Write $u = x/T$. By the quotient rule,

$$\mathrm{b}'(x) = \frac{3x^2(\mathrm{e}^{x/T}-1) - x^3\cdot\frac{1}{T}\mathrm{e}^{x/T}}{(\mathrm{e}^{x/T}-1)^2} = \frac{x^2\left(3\mathrm{e}^{u}-3-u\,\mathrm{e}^{u}\right)}{(\mathrm{e}^{u}-1)^2} = -\frac{x^2\,\mathrm{f}(u)}{(\mathrm{e}^{u}-1)^2}.$$

For $x > 0$ we have $x^2 > 0$ and $(\mathrm{e}^u - 1)^2 > 0$, so $\mathrm{b}'(x) = 0$ exactly when $\mathrm{f}(x/T) = 0$. By the first part this happens only at $x/T = s_0$, so b has a single turning point in $0 < x < \infty$, at

$$x = s_0 T \approx 2.82\,T.$$

Moreover $\mathrm{f}(u) < 0$ for $0 < u < s_0$ and $\mathrm{f}(u) > 0$ for $u > s_0$, so $\mathrm{b}' > 0$ before this point and $\mathrm{b}' < 0$ after it: it is a maximum.

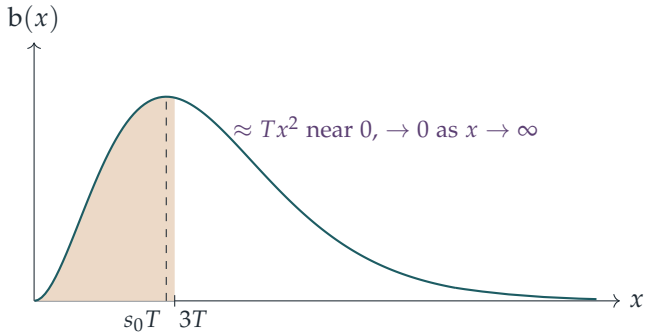
> **Ask yourself** What does b look like near $x = 0$, where both top and bottom vanish?

For small x, $\mathrm{e}^{x/T} - 1 \approx x/T$, so

$$\mathrm{b}(x) \approx \frac{x^3}{x/T} = Tx^2,$$

and the graph leaves the origin like a parabola, with zero gradient. (So $\mathrm{b}(0) = 0$ is the natural value.) For large x the exponential dominates any power: $\mathrm{b}(x) \approx x^3\mathrm{e}^{-x/T} \to 0$, so the x-axis is an asymptote. Since $\mathrm{b}(x) > 0$ for all $x > 0$, the graph rises from the origin to a single hump at $x = s_0 T$ and then decays towards the axis.

Using the first part, the height of the hump is $\mathrm{b}(s_0T) = T^3 s_0^3/(\mathrm{e}^{s_0}-1) \approx 1.42\,T^3$, and $\mathrm{b}(3T) = 27T^3/(\mathrm{e}^3-1)$ is only slightly less.



(The sketch is drawn with $T = 1$; the shaded region is the one used in the final part.)

The form of B

Substitute $x = Tu$, so $\mathrm{d}x = T\,\mathrm{d}u$ and the limits $0, \infty$ are unchanged because $T > 0$:

$$B = \int_0^\infty \frac{T^3u^3}{\mathrm{e}^u - 1}\,T\,\mathrm{d}u = T^4 \int_0^\infty \frac{u^3}{\mathrm{e}^u - 1}\,\mathrm{d}u.$$

The last integral contains no T, so it is a constant K (finite, as we may assume). Hence $B = KT^4$, with $n = 4$.

Estimating K

With the same substitution and $m = 3$,

$$KT^4 = B \approx 2\int_0^{3T} \mathrm{b}(x)\,\mathrm{d}x = 2T^4 \int_0^3 \frac{u^3}{\mathrm{e}^u - 1}\,\mathrm{d}u, \qquad \text{so} \qquad K \approx 2\int_0^3 \frac{u^3}{\mathrm{e}^u - 1}\,\mathrm{d}u.$$

This is twice the shaded area in the sketch (with $T = 1$). Reading heights off the graph, using $\mathrm{e} \approx 2.7$:

$$\mathrm{b}(0) = 0, \quad \mathrm{b}(1) \approx \frac{1}{1.7} \approx 0.59, \quad \mathrm{b}(2) \approx \frac{8}{6.29} \approx 1.27, \quad \mathrm{b}(3) \approx \frac{27}{18.7} \approx 1.45.$$

The trapezium rule with strips of width 1 gives

$$\int_0^3 \approx \tfrac{1}{2}(0) + 0.59 + 1.27 + \tfrac{1}{2}(1.45) \approx 2.58,$$

so

$$K \approx 5.$$

An even rougher estimate treats the shaded region as a triangle of base 3 and height about 1.45, giving area about 2.2 and $K \approx 4.3$. Either way, K is roughly 4 to 5. (The triangle estimate is certainly low, because the curve lies above the straight line from the origin to $(3, \mathrm{b}(3))$.)

Checks

- A numerical bisection puts the root of f at $s_0 \approx 2.8214$, and a numerical scan of b′ with $T = 1.7$ finds exactly one sign change, at $x \approx 4.797 \approx 1.7 s_0$.
- Simpson's rule gives $B/T^4 \approx 6.4939$ for $T = 0.5$, 1 and 2.3: the same value each time, confirming $n = 4$. This matches the known exact value $K = \pi^4/15 \approx 6.494$.
- The integral $\int_0^3 u^3/(\mathrm{e}^u - 1)\,\mathrm{d}u \approx 2.552$, so the approximation the question allows gives $K \approx 5.10$. Our estimate of 5 is about right. The gap from 6.49 is the error in the question's approximation $B \approx 2\int_0^{3T}$, not in our arithmetic.

Common slips

- Forgetting that $\mathrm{f}(0) = 0$: $s = 0$ is a root of f, but it gives $x = 0$, which is outside $0 < x < \infty$, so there is still only one turning point.
- Changing variable in B but forgetting the factor T from $\mathrm{d}x = T\,\mathrm{d}u$, which gives $n = 3$.

Beyond STEP

The function b is Planck's law for black-body radiation, with x playing the role of frequency and T the temperature. The location of the peak, $x = s_0 T$, is Wien's displacement law, and $B \propto T^4$ is the Stefan–Boltzmann law. The exact value $K = \pi^4/15$ comes from expanding $1/(\mathrm{e}^u - 1) = \sum_{k\geq 1} \mathrm{e}^{-ku}$, which gives $K = 6\sum_{k\geq 1} 1/k^4 = 6\zeta(4)$. Justifying that swap of sum and integral is a typical first-year analysis question; see Abbott, *Understanding Analysis*.

Beyond STEP worked solutions: https://beyondstep.keithmcnulty.org/. Solution text CC0 1.0; the question itself is © Cambridge University Press & Assessment and is not reproduced here.